

Util. an imaginary unit used to count satisfaction in examples
Util is the working idea for this part of the chapter.

Cardinal measurement.
measuring satisfaction in numbers, such as 20 utils, so amounts can be added and compared
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The Law of Diminishing Marginal Utility is understood by separating the question into small steps.

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The Law of Diminishing Marginal Utility is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what limitations of the law means before applying it.

The term **Util** means an imaginary unit used to count satisfaction in examples. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): t. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the law of diminishing marginal utility, before moving to the next step.

The term **Cardinal measurement** means measuring satisfaction in numbers, such as 20 utils, so amounts can be added and compared. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): t. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the law of diminishing marginal utility, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in the law of diminishing marginal utility to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

3.3 | LIMITATIONS OF THE LAW | FRAMEWORK | 450

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

THE ELEMENTARY AND PRINCIPLES OF FINANCE



KUTUSH KUMAR, KUTUM PAKHI

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

This section asks what utility and satisfaction means before applying it.

Utility. the satisfaction a person gets from using a good or service
Utility is the working idea for this part of the chapter.

Consumer. a person who buys or uses goods and services to meet wants

Consumer is the working idea for this part of the chapter.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Explain satisfaction as the root of demand and give the first law the reader can test. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): total utility 20, 34, 43, 48, 50; marginal utility 20, 14, 9, 5, 2. A sixth notebook is rated 0 and a seventh would lower satisfaction by 3. Conditions noted on the card: all notebooks of the same kind, bought in a single visit, no change in taste. Notebook price Tk 60 (carried from the shop's price list). The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

3.1 | UTILITY AND SATISFACTION | NEW IDEA | 775

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Utility** means the satisfaction a person gets from using a good or service. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): t. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in utility and satisfaction, before moving to the next step.

The term **Consumer** means a person who buys or uses goods and services to meet wants. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): t. In this chapter, the term matters because it lets the reader state precisely what is

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SUMMARY

Teach the reader how economists argue, what an economic law is, and where microeconomics fits. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

THINK IT THROUGH

2.T1 sort five statements into positive and normative, state a law with its other-things-equal condition, place four decisions under micro or macro

2.T2 a minister says bread should be cheaper — separate the checkable part from the opinion

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

IN HISTORY

The entry directs attention to Great Debasement under Henry VIII, 1544 to 1551, and Thomas Gresham's observation about money; search: Great Debasement Gresham's law. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

REVIEW QUESTIONS

2.3 sort five statements into positive and normative, state a law with its other-things-equal condition, place four decisions under micro or macro

2.4 a minister says bread should be cheaper — separate the checkable part from the opinion

2.5 after the other-things-equal passage and after the micro-macro split

PAUSE AND CHECK 2.P6 after the other-things-equal passage

PAUSE AND CHECK 2.P7 the micro-macro split

The chapter has built one connected chain of reasoning.

A common error is to apply a rule after its condition has changed.

Another error is to confuse a numerical result with the meaning of that result.

The historical case shows the chapter idea in a documented setting.

TO THE READER

This book is written for readers who are meeting the subject for the first time. It assumes no earlier knowledge. Every new word is explained when it first appears, and the book moves slowly wherever an idea is new or difficult.

Beside most paragraphs you will find a note in the narrow column at the edge of the page. A note in italics gives the main point of the paragraph in one line. Read these notes first to see the shape of a chapter, and again when you review.

A word in **bold type** is a new term. It is shown in bold once, where it is first explained, and its meaning is repeated in the narrow column. After that the word appears in ordinary type. Italics are used for emphasis, for foreign words and for the titles of books.¹

Where an everyday word has a special meaning here, a note marked "Word in use" says so. Where an idea is both new and hard, a note reads "Go slowly here", and the text slows down with it.

HOW A CHAPTER RUNS

A chapter opens with a puzzle, then builds its ideas one section at a time. Each section ends with a short recap. The chapter closes with a summary, a problem to think through, the common mistakes people make, a true account from history, and review questions. Questions are numbered, such as 7.1, and their answers are collected at the back of the book.

When a chapter points to another part of the book, it names the chapter and the section, for example Chapter 3, section 3.2. Use the Contents to find it. At the back you will find a glossary of every term, the answers to the questions, a list of further reading, and an index.

This book teaches The Elementary and Principles of Finance one idea at a time, from the very beginning.

The narrow column beside the text holds short notes.

Key term. A new word, shown in bold where it first appears, with its meaning in one line.

A bold word is a new term, and its meaning waits beside it.

Some everyday words have a second, special meaning in this subject.

Each chapter has the same shape, so you always know where you are.

Chapters refer to one another by name, never by page.

Scope, Importance and Objectives is understood by separating the question into small steps.

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The section closes by restating the condition that controls the result.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in scope, importance and objectives is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

Before moving on, keep one question in view: what must remain true for the rule in scope, importance and objectives to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

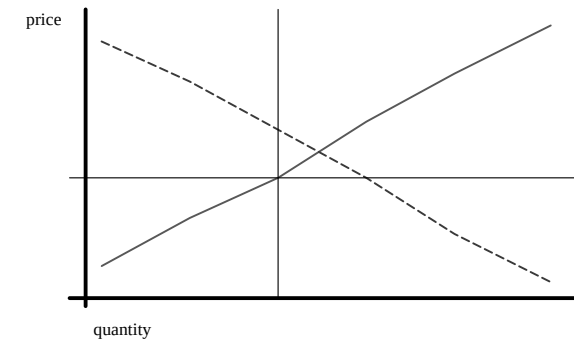


FIG. 2.1. a two-branch map of economics showing the questions that belong to microeconomics and to macroeconomics

The same method remains useful when the surrounding details change.

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

¹ A footnote gives a side point that would interrupt the main argument. Footnotes are numbered within each chapter and appear at the foot of the page.

Ceteris paribus, a Latin phrase meaning other things being equal, used to study one change at a time
Ceteris paribus is the working idea for this part of the chapter.

Economic law, a statement of a regular tendency in human choices, true when other things stay equal

Economic law is the working idea for this part of the chapter.

Economic Laws is understood by separating the question into small steps.

Economic Laws is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what microeconomics and macroeconomics means before applying it.

Model, a simplified picture of a situation that keeps only the features needed to answer a question

Model is the working idea for this part of the chapter.

The term **Ceteris paribus** means a Latin phrase meaning other things being equal, used to study one change at a time. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in economic laws, before moving to the next step.

The term **Economic law** means a statement of a regular tendency in human choices, true when other things stay equal. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in economic laws, before moving to the next step.

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The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

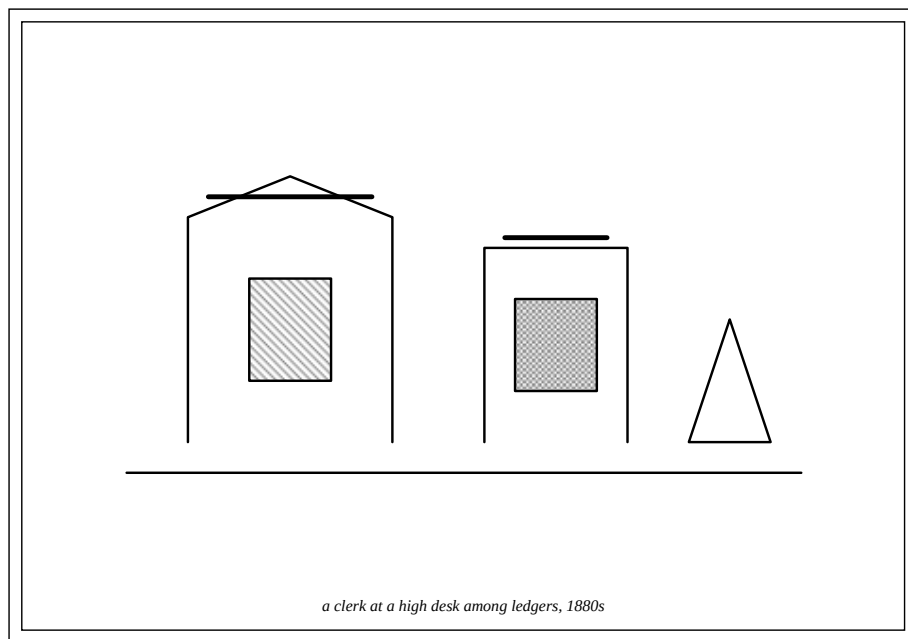
Before moving on, keep one question in view: what must remain true for the rule in economic laws to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

2.3 | MICROECONOMICS AND MACROECONOMICS | FRAMEWORK | 450

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Model** means a simplified picture of a situation that keeps only the features needed to answer a question. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at. In this chapter, the term matters because it lets the reader state precisely what is being measured or



a clerk at a high desk among ledgers, 1880s

CHAPTER 1

Scarcity, Choice and the Economic Problem

What does this chapter let us explain that the earlier ideas could not explain alone?



Show why every society and every shop must choose, and give the reader the first working words of economics. The chapter builds the idea in small stages, then applies it to the running case. The examples keep the stated conditions visible so the reasoning can be checked.

The term **Want** means something a person would like to have, whether or not they can pay for it. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in wants, resources and scarcity, before moving to the next step.

The term **Resource** means anything used to make goods and services: land, labour, capital and enterprise. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in wants, resources and scarcity, before moving to the next step.

The term **Factor of production** means one of the four kinds of resource used in making things: land, labour, capital, enterprise. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in wants, resources and scarcity, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in wants, resources and scarcity to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

Want. something a person would like to have, whether or not they can pay for it

Want is the working idea for this part of the chapter.

Resource. anything used to make goods and services: land, labour, capital and enterprise

Resource is the working idea for this part of the chapter.

Factor of production. one of the four kinds of resource used in making things: land, labour, capital, enterprise

Factor of production is the working idea for this part of the chapter.

Wants, Resources and Scarcity is understood by separating the question into small steps.

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The section closes by restating the condition that controls the result.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

This section asks what science, art and normative views means before applying it.

Science. a body of knowledge built by observing facts, forming explanations and testing them
Science is the working idea for this part of the chapter.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Teach the reader how economists argue, what an economic law is, and where microeconomics fits. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at Tk 60 and 200 copies sell in a week. Ledger note A, in Shirin Meghna's hand: when the price was raised to Tk 66 on 2 March 2026, weekly sales fell to 188. Ledger note B, in Imran Meghna's hand: students should not have to pay more than Tk 60. Note A can be checked against sales records; note B cannot. Regular tendency recorded over three years: textbook sales rise in the two weeks before each term begins (this holds when the shop's prices stay unchanged). Micro and macro: the shop's price and stock are micro matters; the movement of the average price of all goods in Bangladesh is a macro matter. The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

2.1 | SCIENCE, ART AND NORMATIVE VIEWS | NEW IDEA | 550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Science** means a body of knowledge built by observing facts, forming explanations and testing them. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in science, art and normative views, before moving to the next step.

an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in choice and opportunity cost to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

1.3 | THE BASIC PROBLEMS OF AN ECONOMIC ORGANISATION | FRAMEWORK | 550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Opportunity cost** means the value of the best alternative given up when a choice is made. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the basic problems of an economic organisation, before moving to the next step.

The term **Firm** means an organisation that uses resources to make goods or services and sell them. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the basic problems of an economic organisation, before moving to the next step.

The term **Household** means a person or group living together that buys goods and services and supplies resources. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the basic problems of an economic organisation, before moving to the next step.

The term **Economic organisation** means the way a society or business arranges who decides what is made, how and for whom. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state

The section closes by restating the condition that controls the result.

This section asks what the basic problems of an economic organisation means before applying it.

Opportunity cost. the value of the best alternative given up when a choice is made

Opportunity cost is the working idea for this part of the chapter.

Firm. an organisation that uses resources to make goods or services and sell them

Firm is the working idea for this part of the chapter.

Household. a person or group living together that buys goods and services and supplies resources

Household is the working idea for this part of the chapter.

Economic organisation. the way a society or business arranges who decides what is made, how and for whom

Economic organisation is the working idea for this part of the chapter.

automatically. Re-state the problem, identify the new condition, and then apply the rule again.

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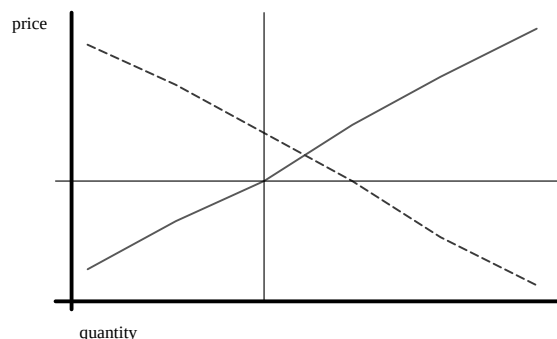


FIG. 1.1. a pair of bars comparing the forgone return of two stock choices, with the chosen option marked

WORKED CASE: THE RUNNING RECORD

Situation. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet; two paid assistants. Record of 12 January 2026: restock fund Tk 400,000 in cash from family savings; shelf space allows one large restock only. Option A, textbooks: purchase cost Tk 400,000; expected sales Tk 560,000; expected sales less purchase cost Tk 160,000. Option B, stationery: purchase cost Tk 400,000; expected sales Tk 540,000; expected sales less purchase cost Tk 140,000. Decision recorded: option A. Opportunity cost of that decision: Tk 140,000. The three problems in shop terms: what to stock (textbooks), how to run the shop (family at the counter plus two assistants), for whom (university students first, schools second). Question. What does the record show when the chapter's central rule is applied to these facts?

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact,

The worked case applies the chapter without adding facts to the record.

The reasoning follows the data rather than an invented story.

The conclusion is limited to what the case supports.

A final check asks whether the result fits the original data.

return to the earlier step rather than adjusting the conclusion to fit the expected answer.

* * *

SUMMARY

Show why every society and every shop must choose, and give the reader the first working words of economics. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

THINK IT THROUGH

1.T1 scarcity versus shortage, forgone value of a choice made from a new price list, which of the three problems a given decision belongs to

1.T2 a student has Tk 9,000 and one free Saturday and must pick between two paid tasks and a course fee — name the opportunity cost

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

IN HISTORY

The entry directs attention to US petrol shortage after the October 1973 oil embargo, with queues and the odd-even plate rationing of 1974; search: 1973 oil embargo gasoline lines odd-even rationing. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

SOURCES

Further reading and historical attribution should be checked against the original works and reliable archival or institutional sources. The chapter does not require the reader to accept a historical interpretation merely because it appears in a secondary account.

REVIEW QUESTIONS

1.3 scarcity versus shortage, forgone value of a choice made from a new price list, which of the three problems a given decision belongs to

1.4 a student has Tk 9,000 and one free Saturday and must pick between two paid tasks and a course fee — name the opportunity cost

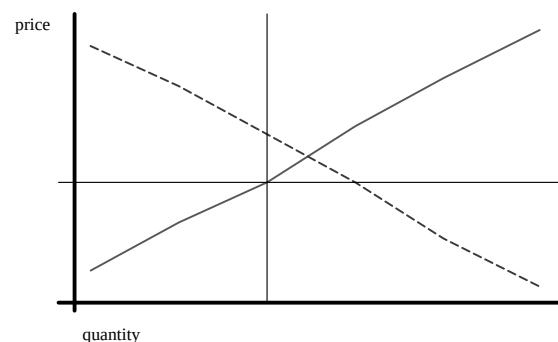


FIG. 3.3. one panel pairing total and marginal utility so the reader sees their link

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

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The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated

The same method remains useful when the surrounding details change.

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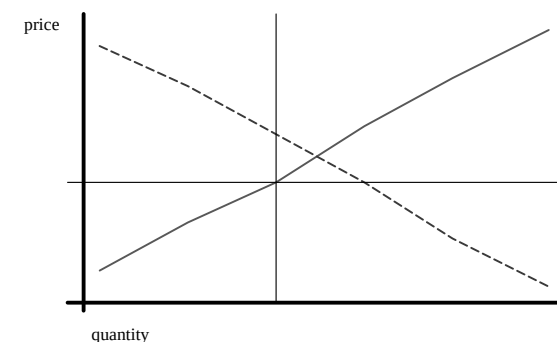


FIG. 6.2. a family of indifference curves, convex to the origin

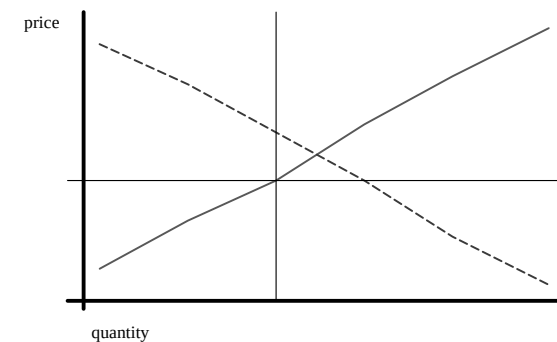


FIG. 6.3. a budget line drawn from a spending plan of Tk 1,200

3.T2 a diner eats from a buffet — find where marginal utility reaches zero and say what a sensible diner does next

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

IN HISTORY

The entry directs attention to The marginal revolution of 1871 to 1874: Jevons, Menger and Walras explain the diamond-and-water puzzle; search: marginal revolution 1871 Jevons Menger diamond water paradox. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

SOURCES

Further reading and historical attribution should be checked against the original works and reliable archival or institutional sources. The chapter does not require the reader to accept a historical interpretation merely because it appears in a secondary account.

REVIEW QUESTIONS

3.3 complete a total-and-marginal table from new scores, state the law with its conditions, give one apparent exception and say why it is not one

3.4 a diner eats from a buffet — find where marginal utility reaches zero and say what a sensible diner does next

3.5 after the total-versus-marginal passage and after the reasons-it-falls passage

PAUSE AND CHECK 3.P6 after the total-versus-marginal passage

PAUSE AND CHECK 3.P7 the reasons-it-falls passage

A common error is to apply a rule after its condition has changed.

Another error is to confuse a numerical result with the meaning of that result.

The historical case shows the chapter idea in a documented setting.

The chapter uses a small number of historical references to anchor its defining ideas.

Indifference map. a set of indifference curves, with curves farther from the origin showing more satisfaction

Indifference map is the working idea for this part of the chapter.

Marginal rate of substitution.

the amount of one good a consumer will give up to gain one more unit of the other

Marginal rate of substitution is the working idea for this part of the chapter.

Indifference Curves and the Budget Line is understood by separating the question into small steps.

Indifference Curves and the Budget Line is understood by separating the question into small steps.

Indifference Curves and the Budget Line is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what consumer equilibrium means before applying it.

Budget line. a line showing every bundle of two goods a consumer can exactly afford with a given income

Budget line is the working idea for this part of the chapter.

The term **Indifference map** means a set of indifference curves, with curves farther from the origin showing more satisfaction. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk 650 and Tk 600; shelf price Tk 650. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in indifference curves and the budget line, before moving to the next step.

The term **Marginal rate of substitution** means the amount of one good a consumer will give up to gain one more unit of the other. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk 650 and Tk 600; shelf price Tk 650. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in indifference curves and the budget line, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in indifference curves and the budget line to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

6.3 | CONSUMER EQUILIBRIUM | TECHNIQUE | 650

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Budget line** means a line showing every bundle of two goods a consumer can exactly afford with a given income. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Show what a buyer gains from a purchase and how a buyer with a fixed budget picks the best bundle. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk 650 and Tk 600; shelf price Tk 650; four buy and one does not. Surpluses: Tk 250, Tk 150, Tk 50, Tk 0, total Tk 450. Choice: a student's monthly spending plan is Tk 1,200 on prescribed books at Tk 150 each and stationery packs at Tk 100 each. Budget line bundles (books, packs): (0, 12), (2, 9), (4, 6), (6, 3), (8, 0). One indifference curve through (4, 6) also passes through (3, 8) and (6, 4); the slope between (3, 8) and (4, 6) is 2 packs per book, between (4, 6) and (6, 4) it is 1 pack per book. Best bundle: (4, 6), cost Tk 600 plus Tk 600 equals Tk 1,200; the price ratio 150 to 100 equals 1.5 packs per book. The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

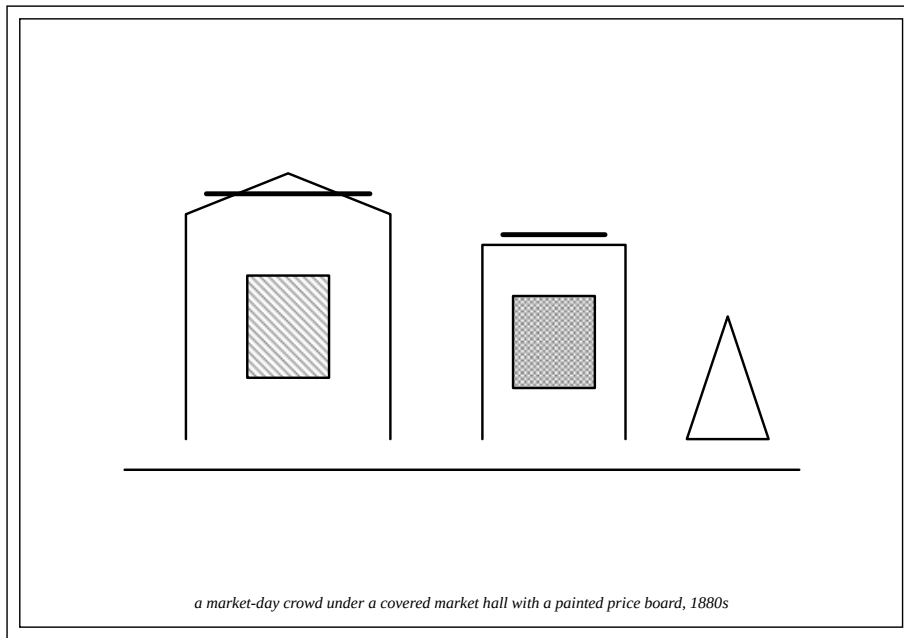
6.1 | CONSUMPTION AND CONSUMER SURPLUS | NEW IDEA | 550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Consumption** means the using up of goods and services to meet wants. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk 650 and Tk 600; shelf price Tk 650. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in consumption and consumer surplus, before moving to the next step.

This section asks what consumption and consumer surplus means before applying it.

Consumption. the using up of goods and services to meet wants
Consumption is the working idea for this part of the chapter.



CHAPTER IV

Demand and the Demand Curve

What does this chapter let us explain that the earlier ideas could not explain alone?



Build the demand idea from a price table to a curve, and show what moves along it and what shifts it. The chapter builds the idea in small stages, then applies it to the running case. The examples keep the stated conditions visible so the reasoning can be checked.

The term **Quantity demanded** means the amount people plan to buy at one particular price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what demand means, before moving to the next step.

The term **Demand schedule** means a table listing the quantity demanded at each of several prices. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what demand means, before moving to the next step.

The term **Demand curve** means a line on a graph showing the quantity demanded at each price, other things equal. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what demand means, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in what demand means to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

4.2 | THE LAW OF DEMAND | DEFINING THEORY | 550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of

Quantity demanded. the amount people plan to buy at one particular price

Quantity demanded is the working idea for this part of the chapter.

Demand schedule. a table listing the quantity demanded at each of several prices

Demand schedule is the working idea for this part of the chapter.

Demand curve. a line on a graph showing the quantity demanded at each price, other things equal

Demand curve is the working idea for this part of the chapter.

What Demand Means is understood by separating the question into small steps.

What Demand Means is understood by separating the question into small steps.

What Demand Means is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what the law of demand means before applying it.

SOURCES

Further reading and historical attribution should be checked against the original works and reliable archival or institutional sources. The chapter does not require the reader to accept a historical interpretation merely because it appears in a secondary account.

REVIEW QUESTIONS

5.3 compute three elasticities from new price-quantity pairs and name each form, say which goods have low elasticity and why, compute income and cross elasticity from new data

5.4 a seller wants more revenue — decide from the elasticity whether to cut or raise the price

5.5 after the formula passage and after the revenue passage

PAUSE AND CHECK 5.P6 after the formula passage

PAUSE AND CHECK 5.P7 the revenue passage

* * *

4.3 | WHAT MOVES THE DEMAND CURVE | NEW IDEA

| 550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Normal good** means a good for which demand rises when buyers' incomes rise. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what moves the demand curve, before moving to the next step.

The term **Inferior good** means a good for which demand falls when buyers' incomes rise. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what moves the demand curve, before moving to the next step.

The term **Substitute goods** means goods that can replace each other, so a rise in one's price raises demand for the other. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what moves the demand curve, before moving to the next step.

The term **Complementary goods** means goods used together, so a rise in one's price lowers demand for the other. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what moves the demand curve, before moving to the next step.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the

This section asks what what moves the demand curve means before applying it.

Normal good. a good for which demand rises when buyers' incomes rise

Normal good is the working idea for this part of the chapter.

Inferior good. a good for which demand falls when buyers' incomes rise

Inferior good is the working idea for this part of the chapter.

Substitute goods. goods that can replace each other, so a rise in one's price raises demand for the other

Substitute goods is the working idea for this part of the chapter.

Complementary goods. goods used together, so a rise in one's price lowers demand for the other

Complementary goods is the working idea for this part of the chapter.

What Moves the Demand Curve is understood by separating the question into small steps.

What Moves the Demand Curve is understood by separating the question into small steps.

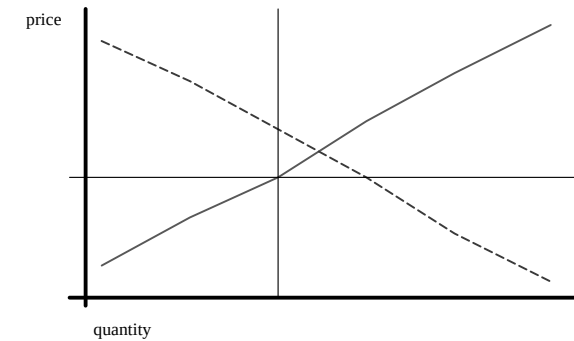


FIG. 5.3. total revenue before and after a price rise for an inelastic and an elastic good

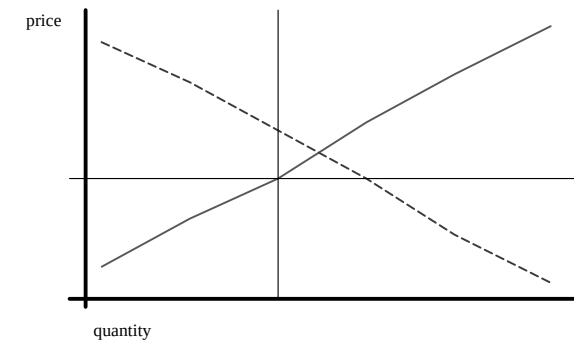


FIG. 5.4. a number line placing income elasticity values for necessities, comforts and luxuries

WORKED CASE: THE RUNNING RECORD

The worked case applies the chapter without adding facts to the record.

Situation. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic); total revenue Tk 12,000 (60 x 200) to Tk 12,408 (66 x 188). Messier rung: midpoint method gives about 0.65. Second good: a Tk 300 atlas, price cut to Tk 270 raised weekly sales from 20 to 30: -10 per cent price, +50 per cent quantity, elasticity 5 in size (elastic); revenue Tk 6,000 to Tk 8,100. Income elasticity: a household income rose from Tk 30,000 to Tk 33,000 a month and its novels bought rose from 50 to 60 a year: +10 per cent and +20 per cent, elasticity 2. Cross elasticity: a rival shop raised its notebook price from Tk 50 to Tk 55 (+10 per cent) and Meghna and Sons sales of its own notebooks rose from 100 to 108 a week (+8 per cent), cross elasticity 0.8. Question. What does the record show when the chapter's central rule is applied to these facts?

result is a description that can be checked against the figures or the stated situation.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

Before moving on, keep one question in view: what must remain true for the rule in related goods and exceptions to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

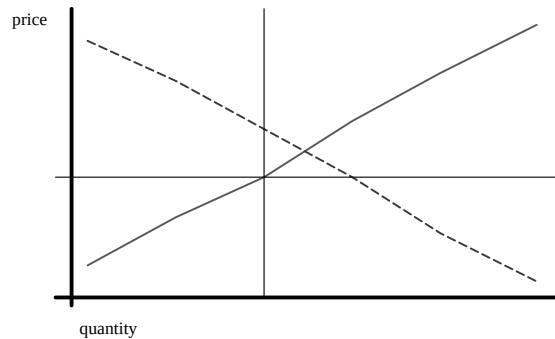


FIG. 4.1. a demand curve drawn from the exercise-book schedule

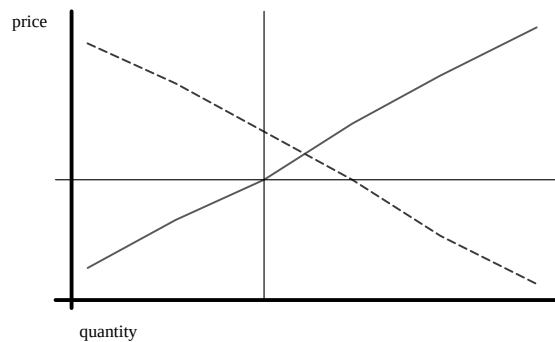


FIG. 4.2. a movement along the curve set beside a shift of the whole curve

Related Goods and Exceptions is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

5.4 | INCOME AND CROSS ELASTICITY | TECHNIQUE | 550

This section asks what income and cross elasticity means before applying it.

Income elasticity of demand. the percentage change in quantity demanded divided by the percentage change in buyers' income

Income elasticity of demand is the working idea for this part of the chapter.

Cross elasticity of demand. the percentage change in quantity demanded of one good divided by the percentage change in another good's price

Cross elasticity of demand is the working idea for this part of the chapter.

Income and Cross Elasticity is understood by separating the question into small steps.

Income and Cross Elasticity is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Income elasticity of demand** means the percentage change in quantity demanded divided by the percentage change in buyers' income. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in income and cross elasticity, before moving to the next step.

The term **Cross elasticity of demand** means the percentage change in quantity demanded of one good divided by the percentage change in another good's price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in income and cross elasticity, before moving to the next step.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in income and cross elasticity is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

Before moving on, keep one question in view: what must remain true for the rule in income and cross elasticity to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

What does the record show when the chapter's central rule is applied to these facts?

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

* * *

SUMMARY

Build the demand idea from a price table to a curve, and show what moves along it and what shifts it. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

THINK IT THROUGH

4.T1 draw a curve from a new schedule, tell a movement from a shift in five events, say how a pen-pack price change affects notebook demand

4.T2 when the price of petrol rises, say what probably happens to demand for natural gas and why

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

IN HISTORY

The entry directs attention to Gregory King's table of harvest shortfalls and wheat prices, published by Charles Davenant in 1699; search: Gregory King Davenant 1699 harvest wheat price. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

The reasoning follows the data rather than an invented story.

The conclusion is limited to what the case supports.

A final check asks whether the result fits the original data.

The chapter has built one connected chain of reasoning.

A common error is to apply a rule after its condition has changed.

Another error is to confuse a numerical result with the meaning of that result.

The historical case shows the chapter idea in a documented setting.

Elastic demand. demand for which the percentage fall in quantity is greater than the percentage rise in price

Elastic demand is the working idea for this part of the chapter.

Inelastic demand. demand for which the percentage fall in quantity is smaller than the percentage rise in price

Inelastic demand is the working idea for this part of the chapter.

Unit elastic demand. demand for which the percentage change in quantity equals the percentage change in price

Unit elastic demand is the working idea for this part of the chapter.

Forms of Price Elasticity is understood by separating the question into small steps.

Forms of Price Elasticity is understood by separating the question into small steps.

Forms of Price Elasticity is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

The term **Elastic demand** means demand for which the percentage fall in quantity is greater than the percentage rise in price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in forms of price elasticity, before moving to the next step.

The term **Inelastic demand** means demand for which the percentage fall in quantity is smaller than the percentage rise in price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in forms of price elasticity, before moving to the next step.

The term **Unit elastic demand** means demand for which the percentage change in quantity equals the percentage change in price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in forms of price elasticity, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

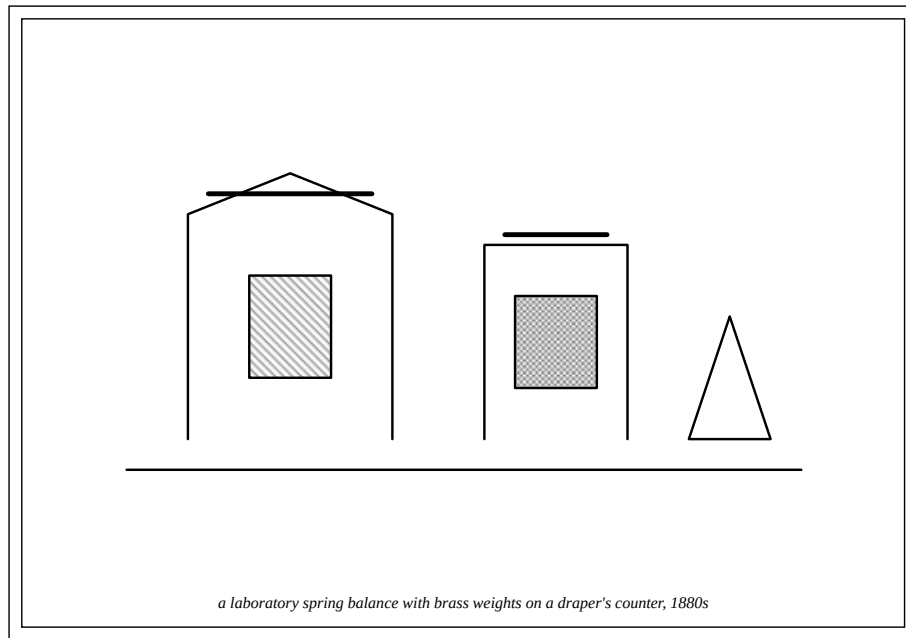
Before moving on, keep one question in view: what must remain true for the rule in forms of price elasticity to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.



CHAPTER V

Elasticity of Demand

What does this chapter let us explain that the earlier ideas could not explain alone?



Teach how to measure how strongly buyers respond, with percentages, and what the answer means for revenue. The chapter builds the idea in small stages, then applies it to the running case.

The examples keep the stated conditions visible so the reasoning can be checked.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Teach how to measure how strongly buyers respond, with percentages, and what the answer means for revenue. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic); total revenue Tk 12,000 (60 x 200) to Tk 12,408 (66 x 188). Messier rung: midpoint method gives about 0.65. Second good: a Tk 300 atlas, price cut to Tk 270 raised weekly sales from 20 to 30: -10 per cent price, +50 per cent quantity, elasticity 5 in size (elastic); revenue Tk 6,000 to Tk 8,100. Income elasticity: a household income rose from Tk 30,000 to Tk 33,000 a month and its novels bought rose from 50 to 60 a year: +10 per cent and +20 per cent, elasticity 2. Cross elasticity: a rival shop raised its notebook price from Tk 50 to Tk 55 (+10 per cent) and Meghna and Sons sales of its own notebooks rose from 100 to 108 a week (+8 per cent), cross elasticity 0.8. The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

5.1 | MEASURING RESPONSIVENESS | TECHNIQUE | 775

This section asks what measuring responsiveness means before applying it.

Elasticity. a measure of how strongly one quantity responds when another quantity changes

Elasticity is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Elasticity** means a measure of how strongly one quantity responds when another quantity changes. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in measuring responsiveness, before moving to the next step.

SUMMARY

Show what a buyer gains from a purchase and how a buyer with a fixed budget picks the best bundle. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

THINK IT THROUGH

6.T1 compute consumer surplus from a new price list, draw a budget line from new prices, say why indifference curves do not cross

6.T2 a student's income doubles with prices unchanged — describe the new budget line and the likely best bundle

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

IN HISTORY

The entry directs attention to Jules Dupuit, 1844, on measuring the usefulness of public works such as bridge tolls; search: Dupuit 1844 utility of public works bridge toll. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

SOURCES

Further reading and historical attribution should be checked against the original works and reliable archival or institutional sources. The chapter does not require the reader to accept a historical interpretation merely because it appears in a secondary account.

REVIEW QUESTIONS

6.3 compute consumer surplus from a new price list, draw a budget line from new prices, say why indifference curves do not cross

6.4 a student's income doubles with prices unchanged — describe the new budget line and the likely best bundle

6.5 after the first indifference curve is drawn and after the tangency passage

PAUSE AND CHECK 6.P6 after the first indifference curve is drawn

PAUSE AND CHECK 6.P7 the tangency passage

The chapter has built one connected chain of reasoning.

A common error is to apply a rule after its condition has changed.

Another error is to confuse a numerical result with the meaning of that result.

The historical case shows the chapter idea in a documented setting.

The chapter uses a small number of historical references to anchor its defining ideas.

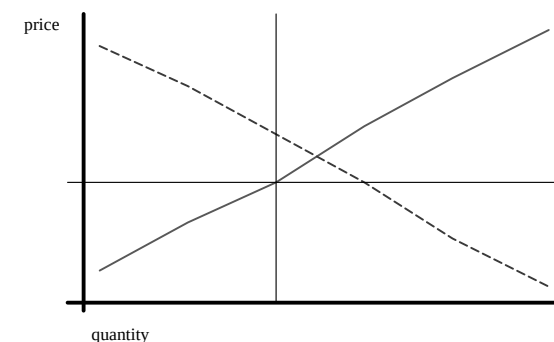


FIG. 9.2. average and marginal product on one diagram showing where they meet

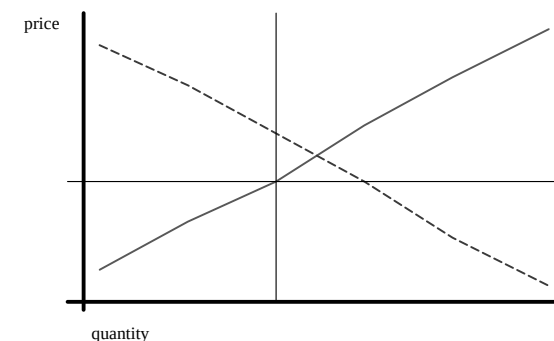


FIG. 9.3. a time bar marking the short run and the long run

Diminishing Returns is understood by separating the question into small steps.

Diminishing Returns is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what returns to scale means before applying it.

Returns to scale. the change in output when all inputs are changed by the same proportion

Returns to scale is the working idea for this part of the chapter.

Increasing returns to scale.

output rises by a larger proportion than inputs

Increasing returns to scale is the working idea for this part of the chapter.

Constant returns to scale. output rises by the same proportion as inputs

Constant returns to scale is the working idea for this part of the chapter.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in diminishing returns is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

Before moving on, keep one question in view: what must remain true for the rule in diminishing returns to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

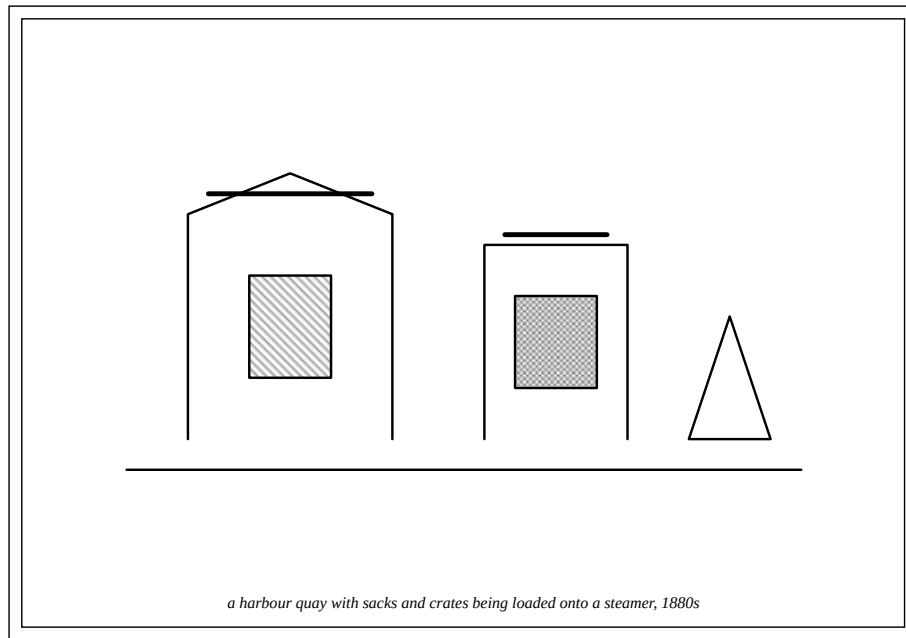
9.4 | RETURNS TO SCALE | TECHNIQUE | 450

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Returns to scale** means the change in output when all inputs are changed by the same proportion. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in returns to scale, before moving to the next step.

The term **Increasing returns to scale** means output rises by a larger proportion than inputs. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in returns to scale, before moving to the next step.

The term **Constant returns to scale** means output rises by the same proportion as inputs. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in returns to scale, before moving to the next step.



CHAPTER VII

Supply and Elasticity of Supply

What does this chapter let us explain that the earlier ideas could not explain alone?



Build the supply idea from a price table to a curve and measure how strongly sellers respond to price. The chapter builds the idea in small stages, then applies it to the running case. The examples keep the stated conditions visible so the reasoning can be checked.

state precisely what is being measured or compared in the law of supply, before moving to the next step.

The term **Supply schedule** means a table listing the quantity supplied at each of several prices. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the law of supply, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

Before moving on, keep one question in view: what must remain true for the rule in the law of supply to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

7.2 | WHAT MOVES THE SUPPLY CURVE | NEW IDEA | 550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Supply curve** means a line on a graph showing the quantity supplied at each price, other things equal. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what moves the supply curve, before moving to the next step.

The term **Law of supply** means when the price of a good rises, quantity supplied rises, and when it falls, quantity supplied falls, other things equal. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300. In this chapter, the term matters because it lets the reader state precisely what is being

Supply schedule. a table listing the quantity supplied at each of several prices

Supply schedule is the working idea for this part of the chapter.

The Law of Supply is understood by separating the question into small steps.

The Law of Supply is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what what moves the supply curve means before applying it.

Supply curve. a line on a graph showing the quantity supplied at each price, other things equal

Supply curve is the working idea for this part of the chapter.

Law of supply. when the price of a good rises, quantity supplied rises, and when it falls, quantity supplied falls, other things equal

Law of supply is the working idea for this part of the chapter.

Short run. a period too short to change at least one input, such as machines or floor space

Short run is the working idea for this part of the chapter.

Long run. a period long enough to change every input

Long run is the working idea for this part of the chapter.

Fixed input. an input whose quantity cannot be changed in the short run

Fixed input is the working idea for this part of the chapter.

Variable input. an input whose quantity can be changed even in the short run

Variable input is the working idea for this part of the chapter.

Short Run and Long Run is understood by separating the question into small steps.

Short Run and Long Run is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

The term **Short run** means a period too short to change at least one input, such as machines or floor space. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in short run and long run, before moving to the next step.

The term **Long run** means a period long enough to change every input. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in short run and long run, before moving to the next step.

The term **Fixed input** means an input whose quantity cannot be changed in the short run. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in short run and long run, before moving to the next step.

The term **Variable input** means an input whose quantity can be changed even in the short run. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in short run and long run, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in short run and long run to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

being measured or compared in exceptions to the law, before moving to the next step.

The term **Elastic supply** means supply for which quantity responds by a larger percentage than price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in exceptions to the law, before moving to the next step.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in exceptions to the law to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

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7.4 | ELASTICITY OF SUPPLY | TECHNIQUE | 775

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Inelastic supply** means supply for which quantity responds by a smaller percentage than price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in elasticity of supply, before moving to the next step.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in elasticity of supply is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is

Elastic supply. supply for which quantity responds by a larger percentage than price

Elastic supply is the working idea for this part of the chapter.

Exceptions to the Law is understood by separating the question into small steps.

Exceptions to the Law is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what elasticity of supply means before applying it.

Inelastic supply. supply for which quantity responds by a smaller percentage than price

Inelastic supply is the working idea for this part of the chapter.

Elasticity of Supply is understood by separating the question into small steps.

Elasticity of Supply is understood by separating the question into small steps.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

This section asks what the production function means before applying it.

Production. the process of turning resources into goods and services

Production is the working idea for this part of the chapter.

Input. a resource used in production

Input is the working idea for this part of the chapter.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Explain how inputs become output, why adding one input yields less and less, and what changes when all inputs grow. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12, 11, 9.6, 8. Long run, scale tables (binds a day): one machine and 2 workers, 40; two machines and 4 workers, 90; four machines and 8 workers, 180; eight machines and 16 workers, 300. Types: first doubling gives 2.25 times output (increasing), second doubling gives 2.0 times (constant), third gives about 1.67 times (decreasing). The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

9.1 | THE PRODUCTION FUNCTION | NEW IDEA | 650

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Production** means the process of turning resources into goods and services. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the production function, before moving to the next step.

The term **Input** means a resource used in production. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14,

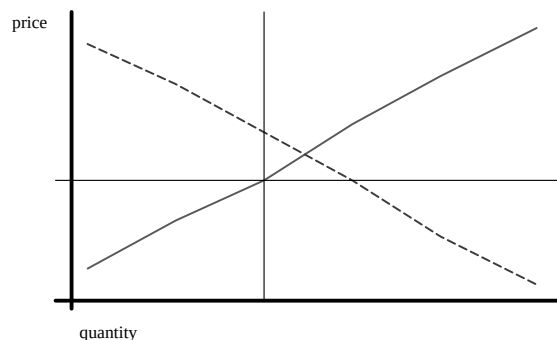


FIG. 7.3. a supply curve of labour that bends backward at high pay

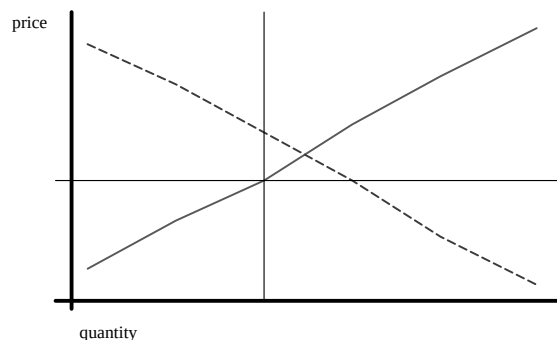


FIG. 7.4. three supply curves of different elasticity through the same point

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated

The same method remains useful when the surrounding details change.

The same method remains useful when the surrounding details change.

REVIEW QUESTIONS

8.3 find equilibrium from new equations, describe the adjustment from a price above or below it, trace the effect of a shift

8.4 a government sets a price ceiling below equilibrium for a staple — describe who gains and who loses

8.5 after the equilibrium solution and after the shortage-and-surplus passage

PAUSE AND CHECK 8.P6 after the equilibrium solution

PAUSE AND CHECK 8.P7 the shortage-and-surplus passage

THINK IT THROUGH

- 7.T1 draw a supply curve from a new schedule, separate five events into movements and shifts, compute and name three elasticities of supply from new data
- 7.T2 explain why a farmer's supply of a perishable crop is inelastic in the short run

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

IN HISTORY

The entry directs attention to The Lancashire cotton famine of 1861 to 1865 and the rise of cotton exports from Egypt and India; search: Lancashire cotton famine 1861 Egypt India cotton exports. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

REVIEW QUESTIONS

- 7.3 draw a supply curve from a new schedule, separate five events into movements and shifts, compute and name three elasticities of supply from new data
- 7.4 explain why a farmer's supply of a perishable crop is inelastic in the short run
- 7.5 after the shift passage and after the elasticity-of-supply formula

PAUSE AND CHECK 7.P6 after the shift passage

PAUSE AND CHECK 7.P7 the elasticity-of-supply formula

A common error is to apply a rule after its condition has changed.

Another error is to confuse a numerical result with the meaning of that result.

The historical case shows the chapter idea in a documented setting.

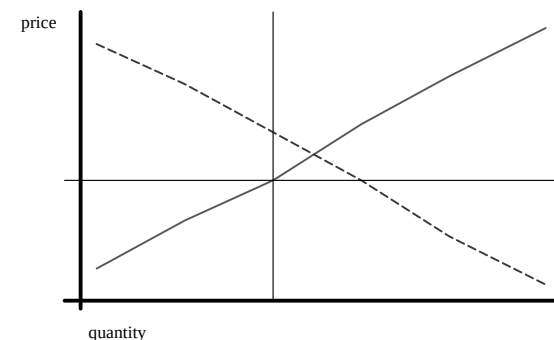


FIG. 8.4. consumer surplus and producer surplus shaded on one diagram

The same method remains useful when the surrounding details change.

The same method remains useful when the surrounding details change.

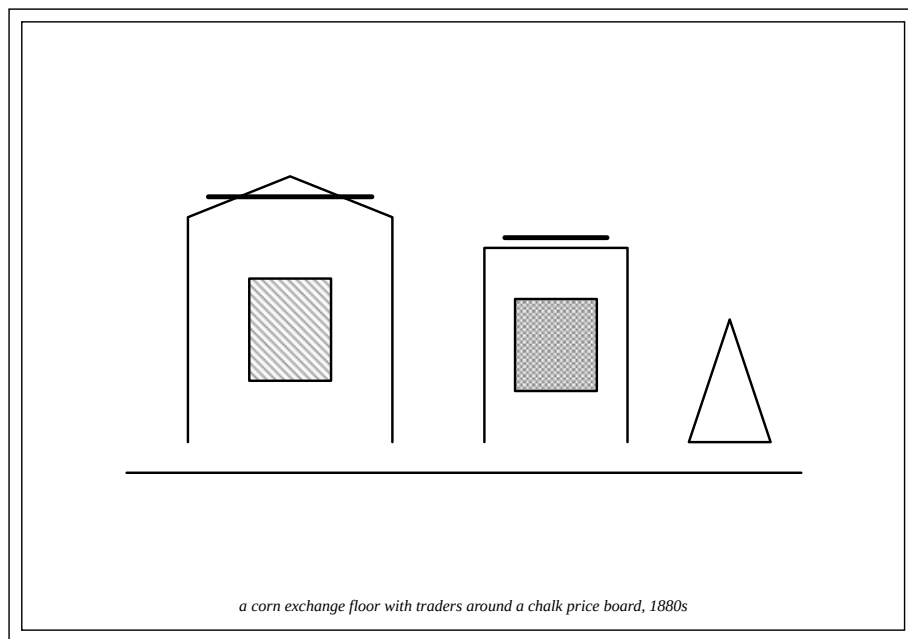
The worked case applies the chapter without adding facts to the record.

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

WORKED CASE: THE RUNNING RECORD

Situation. Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = \text{Tk } 60$, $Q = 900$. Ceiling at Tk 50: $Q_d 1,050$, $Q_s 650$, excess demand 400. Floor at Tk 70: $Q_d 750$, $Q_s 1,150$, excess supply 400. Supply shift: paper cost rises, supply becomes $Q_s = 25P - 760$; new equilibrium $P = \text{Tk } 64$, $Q = 840$. Demand intercept price is Tk 120; supply intercept price is Tk 24 before the shift. Consumer surplus at equilibrium: $0.5 \times (120 - 60) \times 900 = \text{Tk } 27,000$ a week; producer surplus: $0.5 \times (60 - 24) \times 900 = \text{Tk } 16,200$ a week. After the shift: consumer surplus $0.5 \times (120 - 64) \times 840 = \text{Tk } 23,520$; producer surplus $0.5 \times (64 - 30.4) \times 840 = \text{Tk } 14,112$. Substitute-goods link: when the price of petrol rises, demand for natural gas shifts right (kept as a general example, not shop data). Question. What does the record show when the chapter's central rule is applied to these facts?



CHAPTER VIII

How Markets Find a Price

What does this chapter let us explain that the earlier ideas could not explain alone?



Bring demand and supply together to find a price, show why a market is pulled toward it, and measure who gains. The chapter builds the idea in small stages, then applies it to the running case. The examples keep the stated conditions visible so the reasoning can be checked.

Consumer and Producer Surplus in Practice is understood by separating the question into small steps.

Consumer and Producer Surplus in Practice is understood by separating the question into small steps.

Consumer and Producer Surplus in Practice is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

temptation to memorise a formula without understanding what its parts represent.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in consumer and producer surplus in practice is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

Before moving on, keep one question in view: what must remain true for the rule in consumer and producer surplus in practice to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

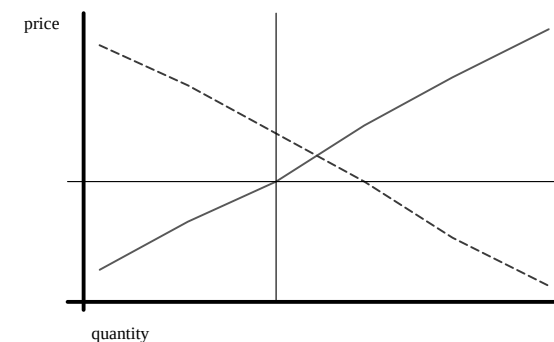


FIG. 8.1. demand and supply curves crossing at the equilibrium point

The term **Equilibrium price** means the price at which quantity demanded equals quantity supplied. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = \text{Tk } 60$, $Q = 900$. Ceiling at Tk 50: $Q_d 1,050$, $Q_s 650$, excess demand 400. Floor at Tk 70:. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in markets and market equilibrium, before moving to the next step.

The term **Equilibrium quantity** means the quantity bought and sold at the equilibrium price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = \text{Tk } 60$, $Q = 900$. Ceiling at Tk 50: $Q_d 1,050$, $Q_s 650$, excess demand 400. Floor at Tk 70:. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in markets and market equilibrium, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

Before moving on, keep one question in view: what must remain true for the rule in markets and market equilibrium to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

8.2 | FORCES TOWARD EQUILIBRIUM | NEW IDEA | 550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Excess demand** means the amount by which quantity demanded exceeds quantity supplied at a price below equilibrium. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = \text{Tk } 60$, $Q = 900$. Ceiling at Tk 50: $Q_d 1,050$, $Q_s 650$, excess demand 400. Floor at Tk 70:. In this chapter, the term matters because it lets the reader state precisely what is being

Equilibrium price. the price at which quantity demanded equals quantity supplied

Equilibrium price is the working idea for this part of the chapter.

Equilibrium quantity. the quantity bought and sold at the equilibrium price

Equilibrium quantity is the working idea for this part of the chapter.

Markets and Market Equilibrium is understood by separating the question into small steps.

Markets and Market Equilibrium is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what forces toward equilibrium means before applying it.

Excess demand. the amount by which quantity demanded exceeds quantity supplied at a price below equilibrium

Excess demand is the working idea for this part of the chapter.

measured or compared in forces toward equilibrium, before moving to the next step.

The term **Excess supply** means the amount by which quantity supplied exceeds quantity demanded at a price above equilibrium. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = \text{Tk } 60$, $Q = 900$. Ceiling at Tk 50: $Q_d 1,050$, $Q_s 650$, excess demand 400. Floor at Tk 70:. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in forces toward equilibrium, before moving to the next step.

The term **Price ceiling** means a legal maximum price that may be charged for a good. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = \text{Tk } 60$, $Q = 900$. Ceiling at Tk 50: $Q_d 1,050$, $Q_s 650$, excess demand 400. Floor at Tk 70:. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in forces toward equilibrium, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in forces toward equilibrium to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

8.3 | SHIFTS AND NEW EQUILIBRIA | TECHNIQUE | 450

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Price floor** means a legal minimum price that may be charged for a good. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = \text{Tk } 60$, $Q = 900$. Ceiling at Tk 50: $Q_d 1,050$, $Q_s 650$, excess demand 400. Floor at Tk

Excess supply. the amount by which quantity supplied exceeds quantity demanded at a price above equilibrium

Excess supply is the working idea for this part of the chapter.

Price ceiling. a legal maximum price that may be charged for a good

Price ceiling is the working idea for this part of the chapter.

Forces Toward Equilibrium is understood by separating the question into small steps.

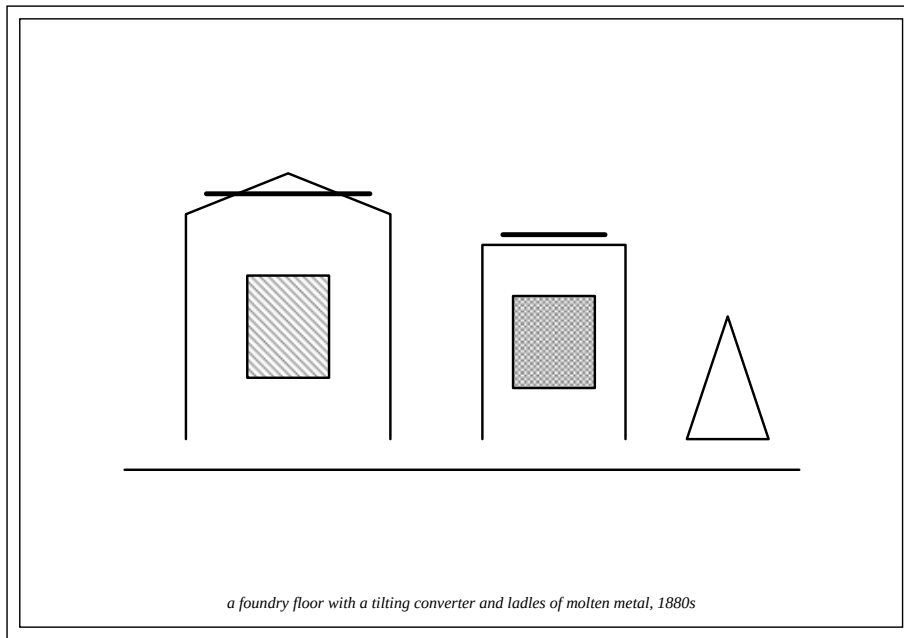
Forces Toward Equilibrium is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what shifts and new equilibria means before applying it.

Price floor. a legal minimum price that may be charged for a good

Price floor is the working idea for this part of the chapter.



CHAPTER X

Costs of Production

What does this chapter let us explain that the earlier ideas could not explain alone?



Show what costs a firm counts, how they change with output, and how to reach a given output at the lowest cost. The chapter builds the idea in small stages, then applies it to the running case. The examples keep the stated conditions visible so the reasoning can be checked.

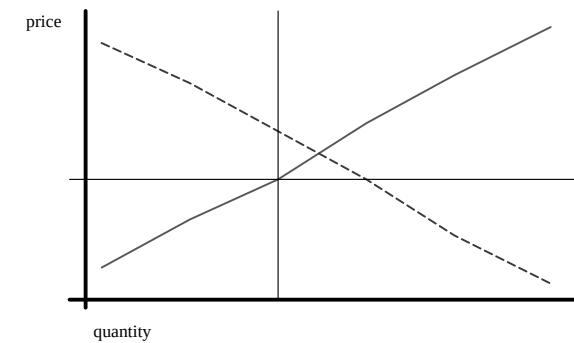


FIG. 12.3. a loanable-funds diagram setting the rate of interest

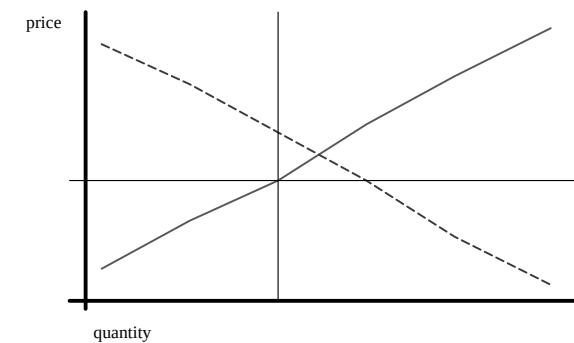


FIG. 12.4. a stacked bar splitting the shop's yearly sales into costs, normal profit and economic profit

WORKED CASE: THE RUNNING RECORD

The worked case applies the chapter without adding facts to the record.

Situation. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000; nil. The shop itself pays Tk 60,000 a month in rent (carried base fact). Wages: shop assistant's nominal wage Tk 18,000 a month; general prices rose 8 per cent in the year; real wage about Tk 16,667 in last year's prices. Interest: a loan of Tk 200,000 at 11 per cent a year costs Tk 22,000 a year. Profit: yearly sales Tk 6,000,000; explicit costs Tk 5,400,000; accounting profit Tk 600,000. Implicit costs: owners' unpaid time Tk 360,000 and return on their own capital Tk 90,000, total Tk 450,000. Economic profit Tk 150,000; normal profit Tk 450,000. Question. What does the record show when the chapter's central rule is applied to these facts?

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures

The reasoning follows the data rather than an invented story.

what is being measured or compared in cost concepts, before moving to the next step.

The term **Explicit cost** means a cost paid out in money to others, such as rent or wages. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in cost concepts, before moving to the next step.

The term **Implicit cost** means the value of resources the owner supplies, such as own time, for which no payment is made. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in cost concepts, before moving to the next step.

The term **Fixed cost** means a cost that stays the same however much is produced in the short run. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in cost concepts, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in cost concepts to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

Explicit cost. a cost paid out in money to others, such as rent or wages

Explicit cost is the working idea for this part of the chapter.

Implicit cost. the value of resources the owner supplies, such as own time, for which no payment is made

Implicit cost is the working idea for this part of the chapter.

Fixed cost. a cost that stays the same however much is produced in the short run

Fixed cost is the working idea for this part of the chapter.

Cost Concepts is understood by separating the question into small steps.

Cost Concepts is understood by separating the question into small steps.

Cost Concepts is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

12.4 | PROFIT | FRAMEWORK | 775

This section asks what profit means before applying it.

Risk bearing. taking on the chance of loss when production is uncertain

Risk bearing is the working idea for this part of the chapter.

Marginal productivity theory. the view that each factor is paid in line with the extra output its last unit adds

Marginal productivity theory is the working idea for this part of the chapter.

Profit is understood by separating the question into small steps.

Profit is understood by separating the question into small steps.

Profit is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Risk bearing** means taking on the chance of loss when production is uncertain. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in profit, before moving to the next step.

The term **Marginal productivity theory** means the view that each factor is paid in line with the extra output its last unit adds. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in profit, before moving to the next step.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in profit is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

Before moving on, keep one question in view: what must remain true for the rule in profit to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in short-run cost curves to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

10.3 | LONG-RUN COST CURVES | NEW IDEA | 225

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Prime cost** means the direct cost of materials and labour that go into a unit of output. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in long-run cost curves, before moving to the next step.

The term **Supplementary cost** means the cost of items apart from prime cost, such as rent and equipment, which remain even when output is nil. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in long-run cost curves, before moving to the next step.

The term **Long-run average cost curve** means a curve showing the lowest average cost of each output when every input can be changed. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in long-run cost curves, before moving to the next step.

The term **Isoquant** means a curve joining the input combinations that all make the same output. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost.

The section closes by restating the condition that controls the result.

This section asks what long-run cost curves means before applying it.

Prime cost. the direct cost of materials and labour that go into a unit of output

Prime cost is the working idea for this part of the chapter.

Supplementary cost. the cost of items apart from prime cost, such as rent and equipment, which remain even when output is nil

Supplementary cost is the working idea for this part of the chapter.

Long-run average cost curve. a curve showing the lowest average cost of each output when every input can be changed

Long-run average cost curve is the working idea for this part of the chapter.

Isoquant. a curve joining the input combinations that all make the same output

Isoquant is the working idea for this part of the chapter.

Real wage. the amount of goods and services a nominal wage can buy

Real wage is the working idea for this part of the chapter.

Interest. the payment for the use of borrowed money or capital over time

Interest is the working idea for this part of the chapter.

Rate of interest. the interest paid in a year as a percentage of the amount borrowed

Rate of interest is the working idea for this part of the chapter.

Wages is understood by separating the question into small steps.

Wages is understood by separating the question into small steps.

Wages is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what interest means before applying it.

The term **Real wage** means the amount of goods and services a nominal wage can buy. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in wages, before moving to the next step.

The term **Interest** means the payment for the use of borrowed money or capital over time. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in wages, before moving to the next step.

The term **Rate of interest** means the interest paid in a year as a percentage of the amount borrowed. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in wages, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in wages to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

12.3 | INTEREST | FRAMEWORK | 225

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the

The central question in least-cost combination is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

Before moving on, keep one question in view: what must remain true for the rule in least-cost combination to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

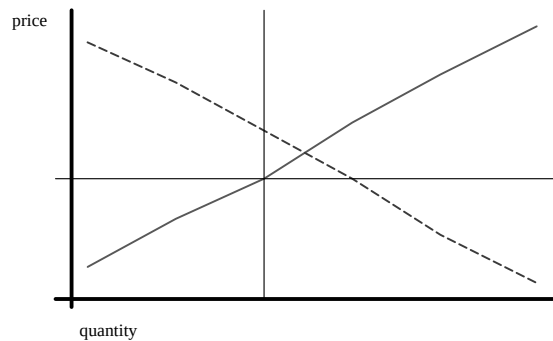


FIG. 10.1. fixed, variable and total cost curves on one diagram

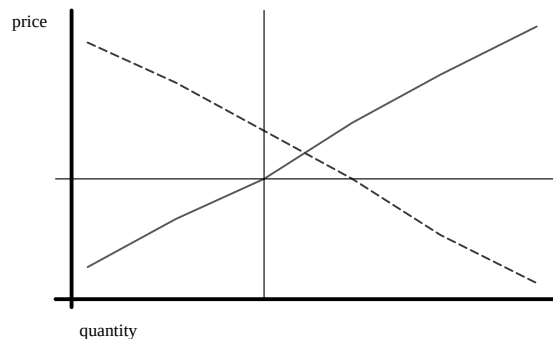


FIG. 10.2. average cost and marginal cost curves, with marginal cost cutting average cost at its lowest point

Least-Cost Combination is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Explain how the money earned from production is shared among land, labour, capital and enterprise. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000; nil. The shop itself pays Tk 60,000 a month in rent (carried base fact). Wages: shop assistant's nominal wage Tk 18,000 a month; general prices rose 8 per cent in the year; real wage about Tk 16,667 in last year's prices. Interest: a loan of Tk 200,000 at 11 per cent a year costs Tk 22,000 a year. Profit: yearly sales Tk 6,000,000; explicit costs Tk 5,400,000; accounting profit Tk 600,000. Implicit costs: owners' unpaid time Tk 360,000 and return on their own capital Tk 90,000, total Tk 450,000. Economic profit Tk 150,000; normal profit Tk 450,000. The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

12.1 | RENT | DEFINING THEORY | 225

This section asks what rent means before applying it.

Factor price. the payment a factor of production earns: rent for land, wages for labour, interest for capital, profit for enterprise

Factor price is the working idea for this part of the chapter.

Economic rent. the payment to a factor above the minimum needed to keep it in its present use

Economic rent is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Factor price** means the payment a factor of production earns: rent for land, wages for labour, interest for capital, profit for enterprise. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in rent, before moving to the next step.

The term **Economic rent** means the payment to a factor above the minimum needed to keep it in its present use. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before

workers and 8 machines Tk 14,400; 24 workers and 2 machines Tk 16,800; least cost Tk 12,000, where marginal product of labour is 30 and of a machine 60 (30 over 600 equals 60 over 1,200). Question. What does the record show when the chapter's central rule is applied to these facts?

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

* * *

SUMMARY

Show what costs a firm counts, how they change with output, and how to reach a given output at the lowest cost. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

THINK IT THROUGH

10.T1 complete the cost columns from new data, find where marginal cost crosses average cost, choose the cheapest input mix from three priced options

10.T2 explain why a family business should count its own unpaid time as a cost when judging success

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

IN HISTORY

The entry directs attention to Henry Bessemer's converter process, 1856, and the fall in the price of steel; search: Bessemer process 1856 steel price fall per ton. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

The reasoning follows the data rather than an invented story.

The conclusion is limited to what the case supports.

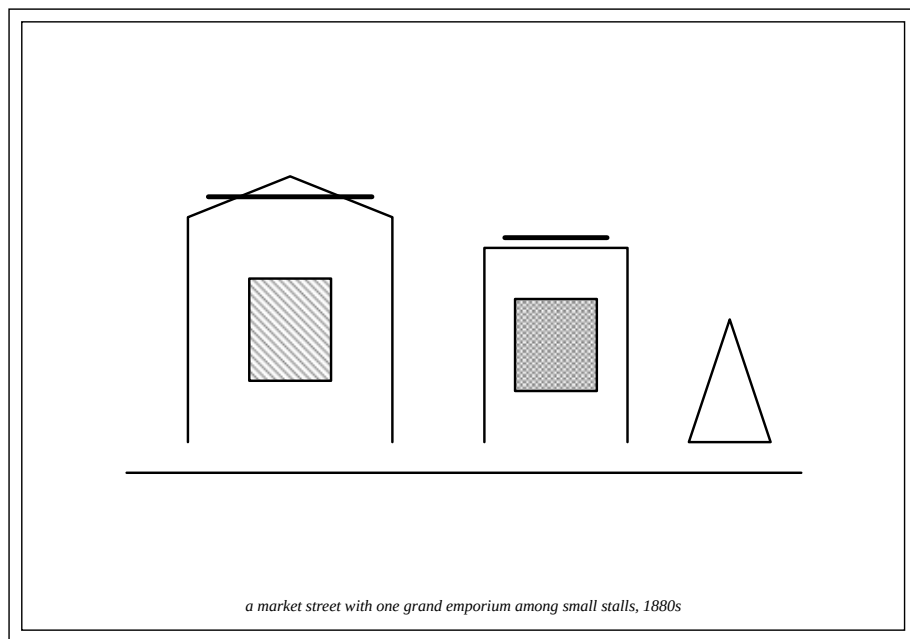
A final check asks whether the result fits the original data.

The chapter has built one connected chain of reasoning.

A common error is to apply a rule after its condition has changed.

Another error is to confuse a numerical result with the meaning of that result.

The historical case shows the chapter idea in a documented setting.



CHAPTER XI

Market Structures and the Firm

What does this chapter let us explain that the earlier ideas could not explain alone?



Compare how firms behave when they face many rivals, one rival or none, and how each structure sets price and output. The chapter builds the idea in small stages, then applies it to the running case.

The examples keep the stated conditions visible so the reasoning can be checked.

The reasoning follows the data rather than an invented story.

The conclusion is limited to what the case supports.

A final check asks whether the result fits the original data.

The chapter has built one connected chain of reasoning.

A common error is to apply a rule after its condition has changed.

Another error is to confuse a numerical result with the meaning of that result.

The historical case shows the chapter idea in a documented setting.

nil. Pricing: price cut from Tk 60 to Tk 55 for the exercise book raised weekly sales from 200 to 210; revenue Tk 12,000 fell to Tk 11,550 (inelastic). Market structure ranking for the shop: exercise books near perfect competition; the sole-authorised reference work a monopoly; mystery novels monopolistic competition. Question. What does the record show when the chapter's central rule is applied to these facts?

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

* * *

SUMMARY

Compare how firms behave when they face many rivals, one rival or none, and how each structure sets price and output. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

THINK IT THROUGH

11.T1 find the profit-maximising output from a new cost table in each of three structures, name the structure from a list of features, decide whether a price cut helps revenue

11.T2 assess the view that firms always aim to maximise profit

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

IN HISTORY

The entry directs attention to The 1911 US Supreme Court decision to break up Standard Oil; search: Standard Oil 1911 Supreme Court dissolution. This account is used only for the historical connection specified by the chapter. It

books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in types of market, before moving to the next step.

The term **Competitive market** means a market with many buyers and sellers, so no single one can set the price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in types of market, before moving to the next step.

The term **Industry** means all the firms that make the same or very similar goods. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in types of market, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

Before moving on, keep one question in view: what must remain true for the rule in types of market to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

11.2 | PERFECT COMPETITION | NEW IDEA | 775

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Price taker** means a firm that must accept the market price because it is too small to change it. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in perfect competition, before moving to the next step.

Competitive market. a market with many buyers and sellers, so no single one can set the price

Competitive market is the working idea for this part of the chapter.

Industry. all the firms that make the same or very similar goods

Industry is the working idea for this part of the chapter.

Types of Market is understood by separating the question into small steps.

Types of Market is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what perfect competition means before applying it.

Price taker. a firm that must accept the market price because it is too small to change it

Price taker is the working idea for this part of the chapter.

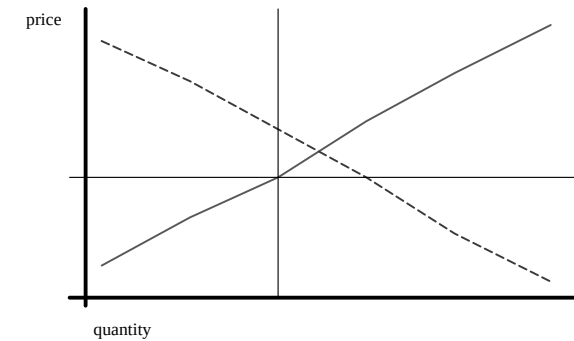


FIG. 11.2. a perfectly competitive firm at the point where marginal cost equals price, with industry beside it

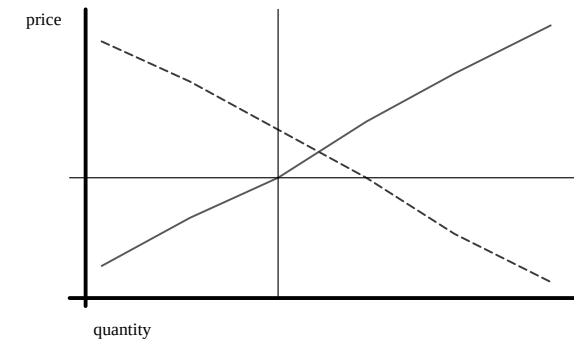


FIG. 11.3. a monopolist's demand and marginal revenue curves with the profit rectangle

what is being measured or compared in monopoly, before moving to the next step.

The term **Pure competition** means a market with many sellers, an identical product and free entry, where buyers may lack full information or free movement. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in monopoly, before moving to the next step.

The term **Monopoly** means a market with one seller and no close substitute, so that seller sets the price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in monopoly, before moving to the next step.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in monopoly to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

11.4 | MONOPOLISTIC COMPETITION | NEW IDEA | 550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Pure monopoly** means a monopoly in which one seller supplies the whole market with a product that has no close substitute. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in monopolistic competition, before moving to the next step.

Pure competition. a market with many sellers, an identical product and free entry, where buyers may lack full information or free movement

Pure competition is the working idea for this part of the chapter.

Monopoly. a market with one seller and no close substitute, so that seller sets the price

Monopoly is the working idea for this part of the chapter.

Monopoly is understood by separating the question into small steps.

Monopoly is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what monopolistic competition means before applying it.

Pure monopoly. a monopoly in which one seller supplies the whole market with a product that has no close substitute

Pure monopoly is the working idea for this part of the chapter.

Bilateral monopoly. a market with a single seller facing a single buyer

Bilateral monopoly is the working idea for this part of the chapter.

Monopolistic competition. a market with many sellers of similar but not identical goods and free entry

Monopolistic competition is the working idea for this part of the chapter.

Monopolistic Competition is understood by separating the question into small steps.

Monopolistic Competition is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what pricing and the firm's goal means before applying it.

Product differentiation. making a firm's good different from rivals' in quality, design, brand or service

Product differentiation is the working idea for this part of the chapter.

The term **Bilateral monopoly** means a market with a single seller facing a single buyer. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in monopolistic competition, before moving to the next step.

The term **Monopolistic competition** means a market with many sellers of similar but not identical goods and free entry. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in monopolistic competition, before moving to the next step.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in monopolistic competition is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

Before moving on, keep one question in view: what must remain true for the rule in monopolistic competition to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

11.5 | PRICING AND THE FIRM'S GOAL | FRAMEWORK | 225

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Product differentiation** means making a firm's good different from rivals' in quality, design, brand or service. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in pricing and the firm's goal, before moving to the next step.

APPENDIX A

Maths You Will Use



A.4 A1.4 | READING A GRAPH | TECHNIQUE | 250

In reading a graph, write the given numbers before choosing an operation. Keep the units visible, carry the arithmetic line by line, and round only at the point required by the question. For a percentage change, compare the change with the starting value. For an average, add the values and divide by their count. For an equation, perform the same operation on both sides. For a graph, read the axes before reading a point.

Given a price of Tk 60 rising to Tk 66, the change is Tk 6. The percentage change is $6 \div 60 \times 100 = 10$ per cent. The important check is that the starting value is 60, not 66. The same discipline applies to ratios, equations and graphs: identify the reference quantity before calculating.

Reading a graph is a tool for checking numerical relationships.

A tiny worked example makes the procedure concrete.

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GLOSSARY

Accounting profit. revenue less explicit costs only. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Art. the skilled use of knowledge to reach a practical aim. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter II*)

Average cost. total cost divided by the number of units produced. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter X*)

Average product. total product divided by the number of units of the variable input. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Backward-bending supply curve. a curve showing that beyond some pay level a worker offers fewer hours as pay rises further. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VII*)

Bandwagon effect. a rise in demand for a good because more people are seen to buy it. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Bilateral monopoly. a market with a single seller facing a single buyer. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Budget line. a line showing every bundle of two goods a consumer can exactly afford with a given income. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VI*)

Cardinal measurement. measuring satisfaction in numbers, such as 20 utils, so amounts can be added and compared. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter III*)

Ceteris paribus. a Latin phrase meaning other things being equal, used to study one change at a time. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter II*)

Choice. picking one option and giving up the others because resources cannot cover every option. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Competitive market. a market with many buyers and sellers, so no single one can set the price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Complementary goods. goods used together, so a rise in one's price lowers demand for the other. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

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shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Economic law. a statement of a regular tendency in human choices, true when other things stay equal. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter II*)

Economic organisation. the way a society or business arranges who decides what is made, how and for whom. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Economic profit. revenue less all costs, explicit and implicit. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Economic rent. the payment to a factor above the minimum needed to keep it in its present use. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Economic system. the set of rules and institutions a country uses to answer its basic economic problems. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Economics. the study of how people, firms and societies choose among uses of limited resources to meet wants. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Elastic demand. demand for which the percentage fall in quantity is greater than the percentage rise in price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter V*)

Elastic supply. supply for which quantity responds by a larger percentage than price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VII*)

Elasticity. a measure of how strongly one quantity responds when another quantity changes. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter V*)

Equilibrium. a state in which nothing pushes the price or quantity to change. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VIII*)

Equilibrium price. the price at which quantity demanded equals quantity supplied. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VIII*)

Equilibrium quantity. the quantity bought and sold at the equilibrium price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VIII*)

Excess demand. the amount by which quantity demanded exceeds quantity supplied at a price below equilibrium. The chapter uses this definition consistently; the running example

Indifference curve. a line joining bundles of two goods that give a consumer the same satisfaction. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VI*)

Indifference map. a set of indifference curves, with curves farther from the origin showing more satisfaction. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VI*)

Industry. all the firms that make the same or very similar goods. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Inelastic demand. demand for which the percentage fall in quantity is smaller than the percentage rise in price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter V*)

Inelastic supply. supply for which quantity responds by a smaller percentage than price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VII*)

Inferior good. a good for which demand falls when buyers' incomes rise. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Input. a resource used in production. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Interest. the payment for the use of borrowed money or capital over time. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Isocost line. a line joining the input combinations that all cost the same total. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter X*)

Isoquant. a curve joining the input combinations that all make the same output. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter X*)

Law of demand. when the price of a good rises, quantity demanded falls, and when it falls, quantity demanded rises, other things equal. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Law of diminishing marginal utility. as a person consumes more units of a good, the extra satisfaction from each further unit falls. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter III*)

Law of diminishing returns. when more of one input is added to fixed inputs, the extra output eventually falls. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Law of supply. when the price of a good rises, quantity supplied rises, and when it falls, quantity supplied falls, other things equal. The chapter uses this

CHAPTER XI · MARKET STRUCTURES AND THE FIRM

11.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

11.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

11.3 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

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11.5 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

11.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

11.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

CHAPTER XII · DISTRIBUTION: RENT, WAGES, INTEREST AND PROFIT

12.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

12.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

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12.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

12.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

Microeconomics. the study of choices made by single households, firms and markets. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter II*)

Model. a simplified picture of a situation that keeps only the features needed to answer a question. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter II*)

Monopolistic competition. a market with many sellers of similar but not identical goods and free entry. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Monopoly. a market with one seller and no close substitute, so that seller sets the price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Nominal wage. the wage counted in money. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Normal good. a good for which demand rises when buyers' incomes rise. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Normal profit. the least profit that keeps a firm in business, equal to the implicit costs of the owner's own resources. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Normative statement. a claim about what ought to be, resting on values and opinion. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter II*)

Oligopoly. a market with a few large sellers, each aware of the others' actions. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Opportunity cost. the value of the best alternative given up when a choice is made. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Ordinal measurement. ranking satisfaction as first, second and third without giving it a number. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter III*)

Output. the goods or services produced. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Percentage change. the size of a change divided by the starting value, multiplied by 100. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter V*)

CHAPTER VII · SUPPLY AND ELASTICITY OF SUPPLY

7.11 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

7.12 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

7.3 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

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7.5 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

7.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

7.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

CHAPTER VIII · HOW MARKETS FIND A PRICE

8.11 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

8.12 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

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8.5 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

8.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

8.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

Production. the process of turning resources into goods and services. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Production function. a rule or table linking the quantities of inputs used to the greatest output they can make. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Profit. the reward of enterprise: what remains of revenue after the other factors are paid. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Profit maximisation. choosing the output at which the gap between total revenue and total cost is greatest. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Pure competition. a market with many sellers, an identical product and free entry, where buyers may lack full information or free movement. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Pure monopoly. a monopoly in which one seller supplies the whole market with a product that has no close substitute. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Quantity demanded. the amount people plan to buy at one particular price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Quantity supplied. the amount sellers plan to offer at one particular price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VII*)

Rate of interest. the interest paid in a year as a percentage of the amount borrowed. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Real wage. the amount of goods and services a nominal wage can buy. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Resource. anything used to make goods and services: land, labour, capital and enterprise. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Returns to scale. the change in output when all inputs are changed by the same proportion. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Risk bearing. taking on the chance of loss when production is uncertain. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Saturation point. the quantity at which marginal utility reaches zero and total utility stops rising. The chapter uses this definition consistently; the running

CHAPTER III · UTILITY AND DIMINISHING MARGINAL UTILITY

3.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

3.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

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3.5 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

3.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

3.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

CHAPTER IV · DEMAND AND THE DEMAND CURVE

4.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

4.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

4.3 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

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4.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

4.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

shows how the term works before it is applied to a changed situation. (*Chapter V*)

Total surplus. consumer surplus plus producer surplus, a measure of the gain from trade. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VIII*)

Total utility. the whole satisfaction from all the units of a good consumed in a period. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter III*)

Unit elastic demand. demand for which the percentage change in quantity equals the percentage change in price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter V*)

Util. an imaginary unit used to count satisfaction in examples. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter III*)

Utility. the satisfaction a person gets from using a good or service. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter III*)

Variable cost. a cost that rises and falls with the amount produced. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter X*)

Variable input. an input whose quantity can be changed even in the short run. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Wage. the payment made to labour for its work. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Want. something a person would like to have, whether or not they can pay for it. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Willingness to pay. the highest price a buyer would accept to get one unit of a good. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VI*)